Feature-space ambiguity, not descriptor collapse, causes residual gross outliers in deep-feature point-cloud displacement estimation: a diagnostic study on a rocky landslide

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Draft v0.1 — code in `investigate/`, branch `feature_descriptor_gedi`

Abstract

Learned local descriptors have made dense 3D displacement estimation between terrestrial laser-scanning (TLS) epochs practical, but on natural, self-similar terrain the resulting vector fields can contain spatially clustered gross outliers that survive downstream geometric filtering. Using the F2S3 pipeline with GeDi descriptors on two epochs of a rocky landslide with moraine-like structures, we diagnose the mechanism behind these outliers. We show that (i) the descriptors do not degenerate — among 861,552 source descriptors, none has a near-duplicate — yet (ii) the raw nearest-neighbour correspondence field is 67% gross outliers, because for a median source point *four* target candidates are within 10% of the best match's feature distance, and these near-tied candidates are scattered across the entire tile (median spatial spread 277 m on a 479 m tile). The ambiguity is therefore deeper than the second neighbour: a top-2 Lowe ratio test is structurally blind to it for the 69% of points with more than two near-tied candidates, and a depth-10 ratio $r_{10} = d_1/d_{10}$ predicts surviving outliers markedly better (AUC = 0.797) than the Lowe ratio (0.636). An offline filter simulation reveals a complementary, practice-relevant result: the best *thresholded* filter is not the best AUC metric — a spatial-spread test with poor AUC (0.607) reduces the raw outlier rate from 67% to 2.4% and, composed with the pipeline's supervoxel filter, yields a 0.47% outlier skeleton, six times cleaner than the strictest ratio test at comparable retention. All conclusions are invariant to the outlier-label threshold over 20–60 m. The study motivates ambiguity-aware rejection in the matching stage and, longer term, descriptors optimised for distinctiveness on self-similar natural surfaces.

Missing for submission: The abstract quantifies results from a single tile of a single scene. Before submission it must either be re-scoped as a case study or be backed by the planned multi-tile / full-resolution replication (Sec. 7).

1 Introduction

Geodetic deformation monitoring increasingly relies on dense 3D displacement vector fields estimated directly between point clouds of two epochs. F2S3 [1] pioneered a feature-based formulation: a deep local descriptor is computed for every point, each source point is assigned its nearest neighbour in *feature space* as corresponding target point, and implausible correspondences are rejected by enforcing local rigid-body consistency inside supervoxels. Unlike surface-differencing methods (C2C, M3C2), this recovers the full 3D displacement vector, including motion tangential to the surface.

The correspondence step, however, contains no notion of *match reliability*: every source point receives a partner, however weak the evidence. On the rocky landslide studied here (Sec. 3), the filtered displacement field is largely coherent (median $|v| \approx 7.6$ m over ∼11 months), but 4.6% of the plotted vectors are gross outliers of tens to hundreds of metres. They are not isolated noise: they form spatially clustered blobs in visually low-texture regions (uniform debris and

moraine-like surfaces), which suggests a systematic matching failure rather than random noise, and they survive the supervoxel filter that should remove them.

Two classical remedies — Lowe's ratio test [4] and a mutual-nearest-neighbour check — examine only the best one or two matches. Before adopting either, we test a sharper hypothesis:

> **H:** On self-similar terrain the descriptor is not sufficiently distinctive: for many source points the *entire* top-k feature neighbourhood — not just the second neighbour — is near-tied, and the near-tied candidates are scattered across the tile. This ambiguity is the mechanism behind the surviving outlier blobs, and a top-2 ratio test can capture it only partially.

From H we derive five questions: (Q1) are the descriptors themselves degenerate, i.e. do many points share near-identical descriptors within one epoch? (Q2) how sharply does the best cross-epoch match stand out from its runners-up? (Q3) where do the near-tied candidates sit *in space*? (Q4) which ambiguity measure best predicts the observed outliers? (Q5) which rejection rule achieves the best density/purity trade-off? Each question maps to one experiment (Sec. 5); a sensitivity analysis verifies that the answers do not depend on the outlier-labelling threshold.

Contributions. (1) A diagnosis separating two failure modes that are often conflated — descriptor *collapse* (many identical descriptors) versus descriptor *indistinctiveness* (distinct descriptors that cannot rank the true partner first) — with evidence that only the latter occurs here. (2) Quantitative evidence that matching ambiguity on natural terrain is *deep* (median four-way near-ties) and *spatially global* (median candidate spread 277 m), which explains why top-2 tests underperform. (3) A methodological caution: filter selection by AUC alone would have discarded the best-performing filter; operating-point (Pareto) analysis is required. (4) A validated, reproducible evaluation protocol for correspondence ambiguity in deep-feature displacement pipelines, including a proxy-label defence and an approximate-search validity check.

2 Related work

Missing for submission: This section has not been researched yet. It should cover, at minimum: (i) point-cloud deformation monitoring baselines (C2C, M3C2, supervoxel-based and piecewise-ICP methods); (ii) learned local 3D descriptors (3DMatch, FCGF, DIP, GeDi, and successors) and their reported distinctiveness properties; (iii) match-reliability tests in 2D and 3D feature matching (ratio test, mutual NN, second-nearest-in-other-cluster, learned match filters such as context normalisation / attention-based filtering); (iv) repetitive-structure ambiguity in place recognition and SLAM (perceptual aliasing), which is the closest analogue to the phenomenon studied here. Currently the bibliography holds only the four works cited directly.

3 Pipeline, data, and labels

3.1 The F2S3 pipeline in brief

F2S3 [1] processes an epoch pair in four stages: (1) voxel-grid downsampling and tiling; (2) per-point deep local feature descriptors — here GeDi [2], a rotation-invariant descriptor producing $D = 32$ dimensions in our configuration; (3) correspondence search: each source descriptor is assigned its (approximate) nearest target descriptor using HNSW [3]; the displacement vector of a source point is the offset to its matched target point; (4) outlier filtering: the source cloud is segmented into supervoxels, a rigid motion is robustly estimated per supervoxel, and correspondences inconsistent with it are discarded. Only stage-(4) survivors are written to the output displacement field.

Table 1: The two displacement fields of the analysed tile. The supervoxel filter turns a two-thirds-outlier raw field into the plotted field; the 4.63% residual outliers are the clustered blobs that motivated this study.

field	points	median $\|v\|$	share $\|v\| \geq 40\,\mathrm{m}$
raw 1-NN	861,552 (100%)	96.1 m	**67.2%**
survivors	189,034 (21.9%)	7.61 m	4.63%

3.2 Scene and run configuration

The test scene ("TCH") is a rocky landslide with moraine-like structures, observed by TLS at two epochs, 2021-09-24 and 2022-08-31 (∼11 months apart). We analyse the spatially subsampled working clouds used during parameter tuning, processed as a single tile with N_s = 861,552 source and N_t = 775,358 target points at 0.2 m voxel size, descriptor neighbourhood radius 6 m, supervoxel radius 4 m. This configuration reproduces the production outlier pattern and is the setting in which the outlier blobs were first observed.

Missing for submission: Scene metadata required for a paper and not derivable from the repository: instrument and scanning geometry, point density of the raw clouds, co-registration procedure and its accuracy between the epochs, and geographic/geological description of the landslide. These must be provided by the operators.

3.3 Two displacement fields, and a data-integrity correction

Two fields must be distinguished throughout:

- **Raw 1-NN field** — the matcher's own output before any filtering: one vector per source point. This is the field the descriptor is directly responsible for.
- **Survivor field** — the subset kept by the supervoxel filter; this is what the pipeline writes out and what the operator plots.

A correction discovered during validation: the output file is *not* row-aligned with the feature matrices — the pipeline writes only surviving points, reordered by supervoxel (189,034 rows against 861,552 source points). Any analysis joining per-point metrics to the output by row index would silently compare unrelated points. We therefore recompute all magnitudes from the correspondence archive (row-aligned by construction) and recover the survivor set by exact 6-D coordinate matching of output rows into correspondence rows, verified to zero magnitude error on all rows.

Table 1 contains the first substantive finding: **the raw matching output is two-thirds gross outliers**. The supervoxel filter is not polishing an essentially correct field — it is rescuing a broken one, and the residual blobs are its leakage.

3.4 Outlier labels: a defended proxy, not ground truth

No independent reference exists for this scene — no GNSS, no manually verified correspondences, no identified stable areas. All "outlier" labels in this study are therefore a *proxy*: a correspondence is labelled a gross outlier when $|v| \geq \theta$, with $\theta = 40\,\mathrm{m}$ as working value. Two observations defend the proxy (Fig. 1):

1. **Bimodality.** The survivor magnitude distribution has a coherent-motion mode (93.3% below 15 m) and a scattered far tail (3.3% beyond 100 m); only 0.66% of survivors fall anywhere between 25 m and 60 m. Any threshold inside that valley labels nearly the same point set.

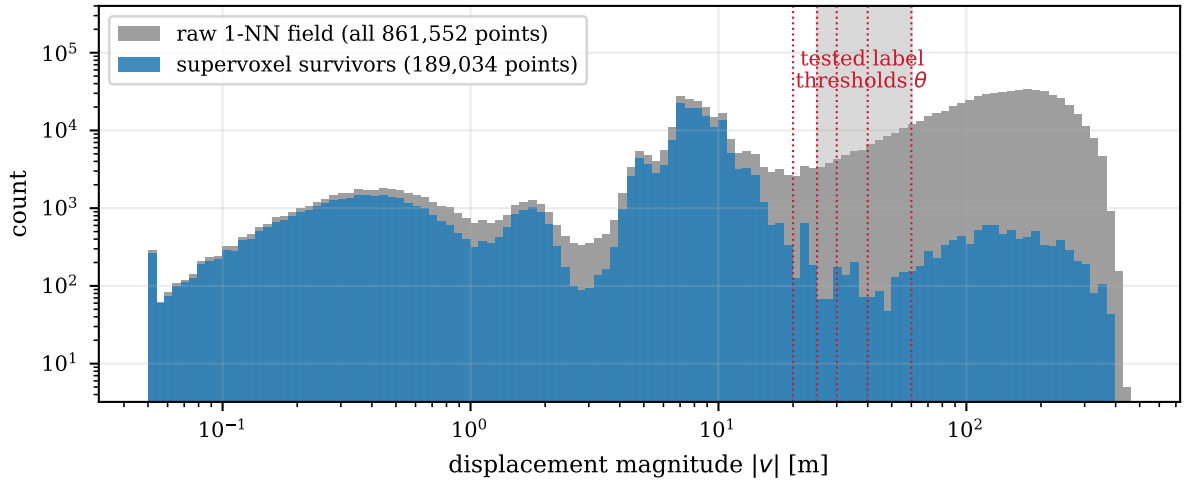


Figure 1: Displacement-magnitude distributions (log–log). The survivor field (blue) is strongly bimodal: coherent motion below ∼15 m and a far outlier tail, separated by a sparsely populated valley (shaded, 25–60 m) in which all tested label thresholds θ (dotted) lie. The raw 1-NN field (grey) is dominated by a broad gross-outlier hump at $|v| > 40$ m.

2. **Threshold invariance.** Every ranking reported below was recomputed at five thresholds θ between 20 and 60 m (Sec. 5.7); no ordering of metrics or filters changes.

What the proxy cannot do is validate individual vectors: a wrong match that lands by chance at a plausible displacement is invisible to it, so all purity numbers are *lower bounds* on the true error rate.

We use four recurring point groups: *raw outliers* ($|v| \geq 40$ m, any point; 578,677 = 67.2% of N_s); *survivors* (kept by the supervoxel filter; 189,034); *trusted* points (survived and $|v| < 15$ m; 176,435); and *residual outliers* (survived and $|v| \geq 40$ m; 8,756 = 4.63% of survivors — the blobs). The *baseline* operating point of the current pipeline is the survivor field: 21.9% density at 4.63% outliers.

3.5 Validity of exact k-NN analysis for the HNSW-based pipeline

All analyses use exact (brute-force, GPU-chunked) k-nearest-neighbour search, while the pipeline uses approximate HNSW. If the two disagreed materially, our metrics would describe a different matcher than the one that produced the field. On this tile the HNSW-chosen partner equals the exact 1-NN for 99.960% of source points (343 disagreements out of 861,552); in every disagreement the HNSW pick lies within the exact top-20, with a median feature-distance excess of 1.5% (95th percentile 8.6%). Exact-kNN statements therefore transfer to the production matcher.

4 Ambiguity metrics

For source point i, let $d_1 \leq d_2 \leq \cdots \leq d_{20}$ denote its sorted L2 distances to the 20 nearest target descriptors (computed once and reused by all experiments).

Lowe ratio. $r_2 = d_1/d_2$ [4]. Values near 1 mean the runner-up is nearly as good as the winner; the classical criterion accepts a match if $r_2 < 0.8$. It sees exactly one competitor.

Deep ratio. $r_{10} = d_1/d_{10}$, and the equivalent flatness $= (d_{10} - d_1)/d_1 = 1/r_{10} - 1$. These are ad-hoc extensions defined for this study (no literature source claimed); being monotone transforms of each other they are identical under rank-based evaluation and we report only r_{10}.

Ambiguity depth. $\text{depth}_\varepsilon = \#\{\, j \le 20 : d_j \le (1+\varepsilon)\, d_1 \,\}$ for $\varepsilon \in \{0.05, 0.1, 0.2\}$. Here ε is a *relative distance tolerance*: $\text{depth}_{0.1}$ counts the candidates within 10% of the winner's distance — how many are "effectively tied for first place". Depth 1 means the winner stands alone; depth $\gg 2$ is precisely the regime a top-2 test cannot represent. There is no canonical ε, so we report three and let the predictive evaluation and the sensitivity analysis judge.

Spatial spread. Map the top-10 candidates to their coordinates and take the maximum pairwise Euclidean distance among them. Spread is the only *geometric* measure: it asks not how close the runner-up *distances* are, but where the alternative matches would put the point.

Within-epoch self-similarity (collapse test). $d_1^{self}(i)$ is the distance from source descriptor i to its nearest *other* source descriptor, and $n_\varepsilon(i)$ counts other source descriptors within an absolute radius $\varepsilon_1 = 0.1 \cdot \text{median}(d_1^{self})$. Unlike the depth tolerance, ε_1 is a single global radius tied to the data scale (one tenth of the typical nearest-neighbour distance), so n_ε counts *near-duplicates* rather than merely close descriptors.

Evaluation score. We score each metric's ability to separate outliers from non-outliers by the area under the ROC curve (AUC) — equivalently, the probability that a randomly drawn outlier has a larger metric value than a randomly drawn non-outlier — computed rank-based with tie correction; 0.5 means no information, 1.0 perfect separation. The AUC is unrelated to the ε parameters: ε shapes a metric's values, AUC then scores those values against the labels of Sec. 3.4.

Missing for submission: No uncertainty quantification yet: AUC values and outlier rates are point estimates. Bootstrap confidence intervals (resampling source points, or better, spatial block bootstrap to respect spatial correlation) should be added.

5 Results

Each subsection states the expectation under hypothesis H before the observation, and flags deviations.

5.1 Q1: No descriptor collapse

Expectation under H: if GeDi degenerates on self-similar surfaces, large regions should show $d_1^{self} \approx 0$ and $n_\varepsilon \gg 0$.

Observation (Fig. 2): the opposite. Median $d_1^{self} = 0.211$, giving $\varepsilon_1 = 0.021$; $n_\varepsilon = 0$ for *every one* of the 861,552 source descriptors — not a single descriptor has even one near-duplicate within its own epoch. GeDi does not collapse on this terrain.

This negative result sharpens H rather than refuting it: the failure mode is not "the network emits the same vector everywhere" but "the vectors, while distinct, do not place the correct partner first by a reliable margin" (Secs. 5.2–5.3). It also predicts, correctly, that n_ε carries no information about outliers (AUC exactly 0.500, Table 2).

5.2 Q2a: The winner barely stands out

Expectation under H: trusted points should show a distance profile that rises steeply after d_1 (a distinct winner), residual outliers a flat profile. In an ideal world the (d_1, r_2) distribution would concentrate at $r_2 \ll 1$ — a clear winner for every point.

Observation (Fig. 3): ambiguity manifests as mass piling up against $r_2 \to 1$ for the bulk of all points. Group means: $r_2 = 0.919$ (trusted) versus 0.951 (residual outliers); $r_{10} = 0.766$ versus 0.858. Two consequences. First, even for *trusted* points the runner-up is on average only

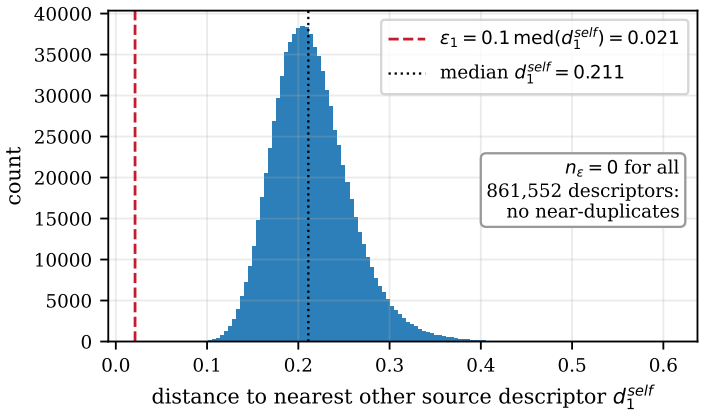


Figure 2: Within-epoch self-similarity of the source descriptors. The distance to the nearest other descriptor is bounded far away from the near-duplicate radius ε_1 (dashed): no descriptor collapse.

~8% worse than the winner — far from the ideal-world picture — so classical Lowe filtering at $r_2 < 0.8$ would discard the overwhelming majority of correct matches on this terrain (it keeps 1.4% of all points, Table 3). Second, the group separation in r_2 (0.032) is one third of the separation in r_{10} (0.092): first evidence that the discriminative signal lives deeper than rank 2.

5.3 Q2b: The ambiguity is deep

Expectation under H: residual outliers should have depth $\gg 2$ far more often than trusted points, and a substantial fraction of all points at depth > 2 would prove a top-2 test structurally blind here.

Observation (Fig. 4): confirmed. At $\varepsilon = 0.1$ the median ambiguity depth is **4**, and **69.3%** of all source points have depth > 2; group means are 2.92 (trusted) versus 5.81 (residual outliers). For over two thirds of the points, the situation a top-2 ratio test models — one winner, one challenger — simply does not exist; there is a cluster of challengers.

Caveat: at $\varepsilon = 0.2$ the median depth equals $20 = k$, i.e. the statistic is right-censored by the top-20 search; true depths at loose tolerance are larger than reported ($k = 100$ would be needed to measure them). This does not affect any ranking below, which uses $\varepsilon = 0.1$.

5.4 Q3: Near-tied candidates are scattered across the tile

Expectation under H: trusted points → small spread (candidates cluster around the true partner); ambiguous points → spread approaching the tile diameter.

Observation (Fig. 5): the second half is confirmed emphatically — median spread of the top-10 candidates is **277 m on a tile with a 479 m diagonal** (90th percentile 343 m). The alternative matches of a typical point are not local jitter around the true partner; they are distributed over the entire scene, which is exactly the mechanism that converts a near-tie in feature space into a hundreds-of-metres blunder in the displacement field.

Deviation from expectation: trusted points *also* show large mean spread (220 m versus 262 m for residual outliers). Global candidate scatter is near-universal on this terrain; what distinguishes reliable points is membership in the small-spread *tail* (spread $<$ 10–20 m), not a low average. This asymmetry explains the seemingly paradoxical behaviour of spread in Secs. 5.5–5.6.

5.5 Q4: Which metric predicts the outliers?

Expectation under H: depth- and spread-type metrics should outrank the top-2 ratio in AUC.

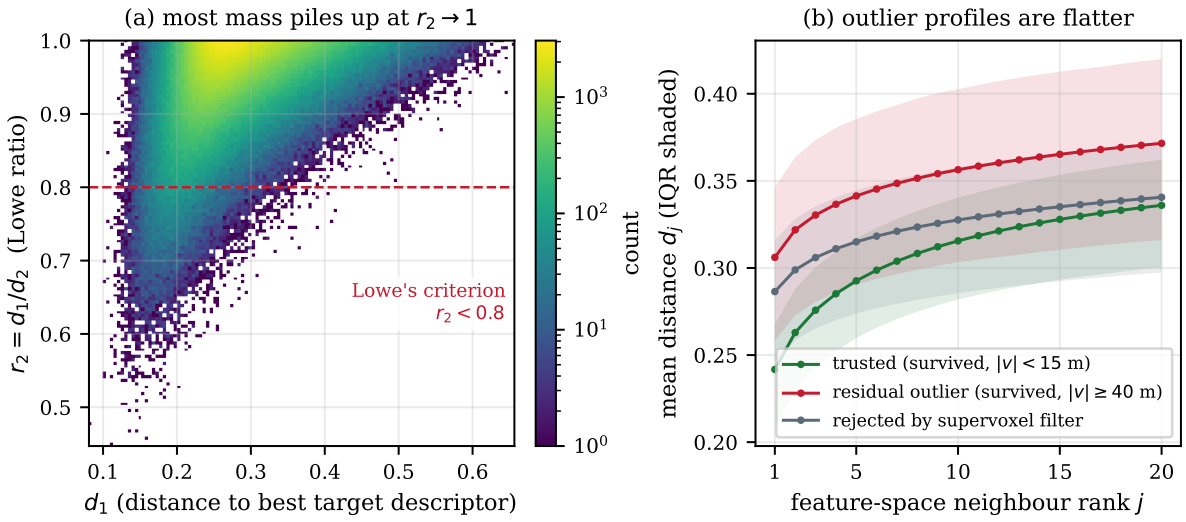


Figure 3: Cross-epoch match quality. (a) Joint density of best-match distance d_1 and Lowe ratio r_2 (log colour scale): most points sit near $r_2 \to 1$, i.e. the runner-up is nearly as good as the winner; Lowe's classical acceptance region lies below the dashed line. (b) Mean top-20 distance profiles by group (bands: interquartile range): residual outliers (red) have flatter profiles than trusted points (green), but the difference is gradual, not categorical.

Table 2: AUC of each ambiguity metric as a predictor of the gross-outlier label ($|v| \geq 40$ m), for the raw target (all source points) and the residual target (supervoxel survivors only). Flatness is omitted: it is a monotone transform of r_{10} with identical AUC.

metric	raw target	residual target
$r_{10} = d_1/d_{10}$	0.781	**0.797**
d_1 (match distance)	0.680	0.791
$\text{depth}_{0.2}$	0.765	0.772
$\text{depth}_{0.1}$	0.712	0.699
$\text{depth}_{0.05}$	0.641	0.629
r_2 (Lowe ratio)	0.640	0.636
spread (top-10)	0.629	0.607
n_ε (collapse)	0.500	0.500

Observation (Table 2, Fig. 6): confirmed for the depth direction: r_{10} (AUC 0.797 on the residual target) decisively outranks r_2 (0.636) — the discriminative information sits between ranks 2 and 10, invisible to a top-2 test. Two unexpected results: (i) the raw match distance d_1 alone is nearly as predictive as r_{10} among survivors (0.791): a large absolute feature distance is itself a strong warning sign; (ii) spread scores second-worst (0.607) — yet Sec. 5.6 shows it is the best *filter*, a contradiction resolved there.

5.6 Q5: Filter operating points and the Pareto view

Every candidate rejection rule is a boolean mask over source points, so its effect can be simulated offline. Because a feature-space filter would run *before* the (unchanged) supervoxel filter, each rule is evaluated as the intersection of both filters and quoted as an *operating point*: retained density (% of all N_s; baseline 21.9%) versus outlier rate among kept points. Rule A *dominates* rule B if it is simultaneously denser and cleaner; the set of non-dominated rules is the *Pareto frontier* — the menu of rational choices — and the "knee" is where sacrificing more density stops buying meaningful purity.

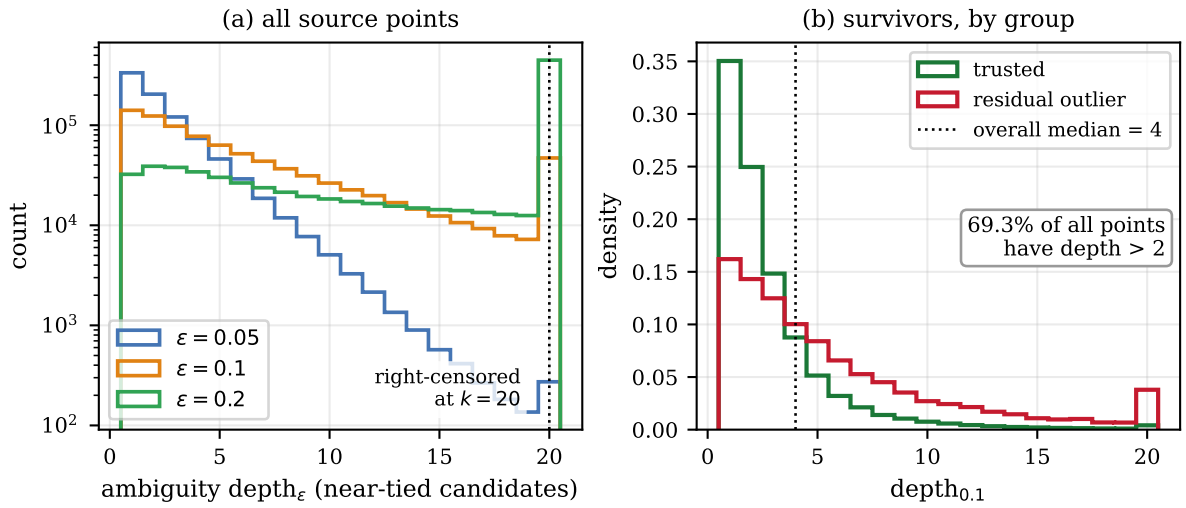


Figure 4: Ambiguity depth. (a) Depth distributions for three tolerance values (log counts); the spike at 20 is censoring by the top-k search, not data. (b) At $\varepsilon = 0.1$, residual outliers (red) are shifted to systematically larger depths than trusted points (green); the overall median is 4.

Expectation under H: a depth- or spread-based rejection should reach a better operating point than the ratio test.

Observation (Table 3, Fig. 7):

1. **Like-for-like, depth beats ratio.** At matched density (12.8% vs 12.5%), depth$_{0.1} \leq 2$ yields 2.42% outliers versus 3.10% for $r_2 < 0.95$. The AUC conclusion carries through to operating points.

2. **Spread is the best single filter by a wide margin, despite its poor AUC.** Acting alone on the raw field it cuts 67.2% → 2.4% outliers while keeping 3.0% of points; composed with the supervoxel filter it reaches 2.1% density at **0.47%** outliers — ∼6× cleaner than the strictest ratio test at comparable retention. Resolution of the AUC contradiction: AUC scores a metric's ranking over its *entire* range, but spread is uninformative over most of its range (Sec. 5.4: nearly everything scatters widely) and nearly pure in its low tail. A low-recall/high-precision signal loses at AUC and wins as a thresholded filter. *Methodological consequence: selecting a filter by AUC alone would have picked r_{10} and discarded the strongest filter.*

3. **Mutual NN alone is insufficient** (50.2% outliers on the raw field): when the descriptor cannot rank the true partner first, the wrong assignment is often symmetric, so the mutuality check passes.

4. The median $|v|$ is stable (7.3–7.8 m) under every rule: the central motion estimate was never in question; the rules differ only in how much of the tail they admit.

5. No rule dominates the baseline: every gain in purity costs density. The choice of knee depends on downstream use (Sec. 8).

5.7 Robustness to the label threshold

Recomputing base rates, all metric AUCs, and all filter outlier rates at $\theta \in \{20, 25, 30, 40, 60\}$ m changes no ordering anywhere in the range (Fig. 8; e.g. spread < 10 m moves from 0.76% to 0.43% outliers monotonically). Combined with the bimodality of Fig. 1, the proxy label is as solid as it can be made without reference data.

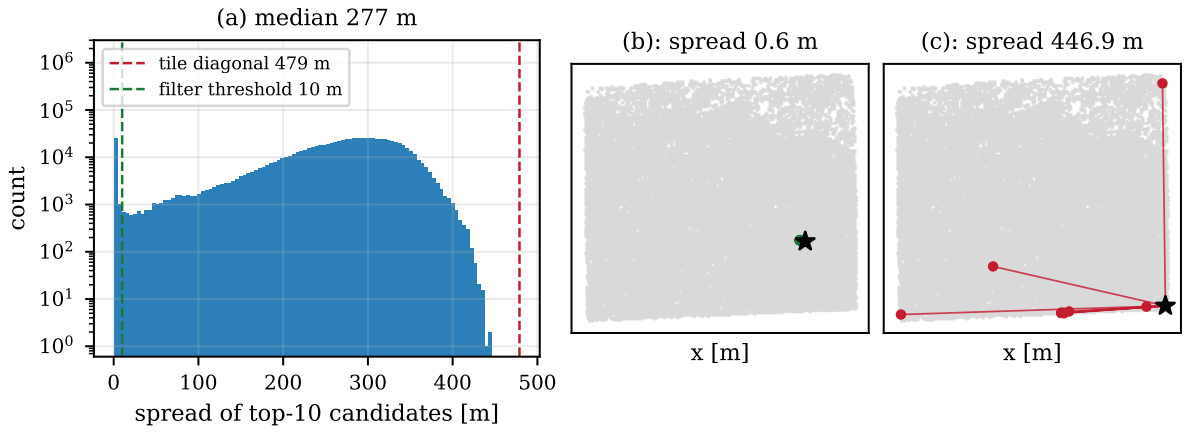


Figure 5: Spatial spread of the top-10 candidate sets. (a) Distribution (log counts): the mass sits at hundreds of metres, with a distinct low-spread spike near zero. (b) A distinctive point: all 10 candidates coincide at the source location (green dots hidden behind the star). (c) An ambiguous point: near-tied candidates span the full tile. Grey: target cloud (subsampled), top view.

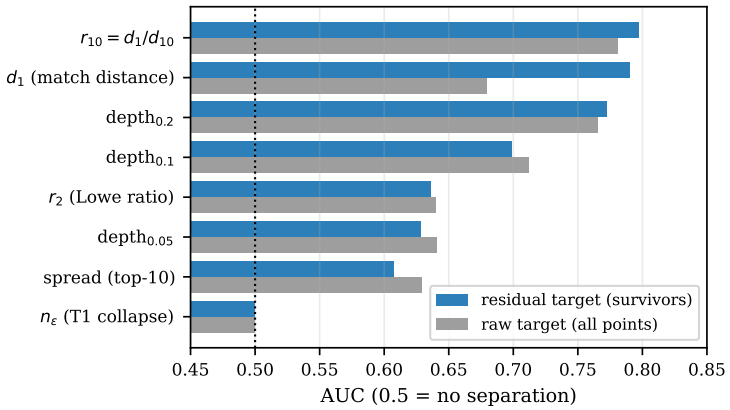


Figure 6: Separability (AUC) of the ambiguity metrics for both targets, sorted by the residual target. The collapse count n_ε is exactly uninformative, consistent with Sec. 5.1.

6 Discussion

The hypothesis, revisited. H is confirmed in its mechanism, refuted in one detail, and exceeded in one respect. *Confirmed:* the surviving outlier blobs arise from feature-space ambiguity that is deeper than the second neighbour (median four-way near-ties; 69% of points beyond the reach of a top-2 test; $r_{10} \gg r_2$ in predictive power), and the near-tied candidates scatter across the full tile, which converts ambiguity into large blunders. *Refuted detail:* there is no descriptor collapse; the descriptors are distinct but insufficiently separated to rank reliably. *Exceeded:* the raw 1-NN field is 67% gross outliers — the problem is not a blemish on a good matching stage; the supervoxel filter currently carries the pipeline.

No filter adds information. Because the descriptors are individually distinct (Sec. 5.1) yet collectively ambiguous, rejection tests can only *remove* unreliable points, never recover correct partners the descriptor failed to rank first. Every operating point in Fig. 7 trades density for purity; none dominates the baseline. Genuine improvement requires either additional evidence at matching time (spatial regularisation, geometric verification of candidate sets — the spread result shows the geometry of the candidate set carries information the feature distances do not) or a more distinctive descriptor on this terrain class.

Table 3: Filter operating points (selection; full grid in `investigate/out/t6_pareto_tile_000.csv`). "Kept" is the density after composing the rule with the supervoxel filter; "raw alone" is the outlier rate of the rule applied to the raw 1-NN field without the supervoxel filter.

filter	kept [% of N_s]	outliers [%]	median $\lvert v\rvert$ [m]	raw alone [%]
none (baseline)	21.9	4.63	7.61	67.2
$\text{depth}_{0.1} \leq 3$	16.0	2.73	7.61	49.8
$\text{depth}_{0.1} \leq 2$	12.8	2.42	7.62	46.3
$r_2 < 0.95$	12.5	3.10	7.62	54.8
mutual NN	8.6	1.54	7.49	50.2
$r_2 < 0.80$ (Lowe's value)	1.4	0.67	7.78	14.6
spread $< 10\,\text{m}$	2.1	**0.47**	7.32	**2.4**
mutual $\wedge$ $r_2 < 0.9$ $\wedge$ spread$< 10\,\text{m}$	0.5	0.13	7.31	0.6

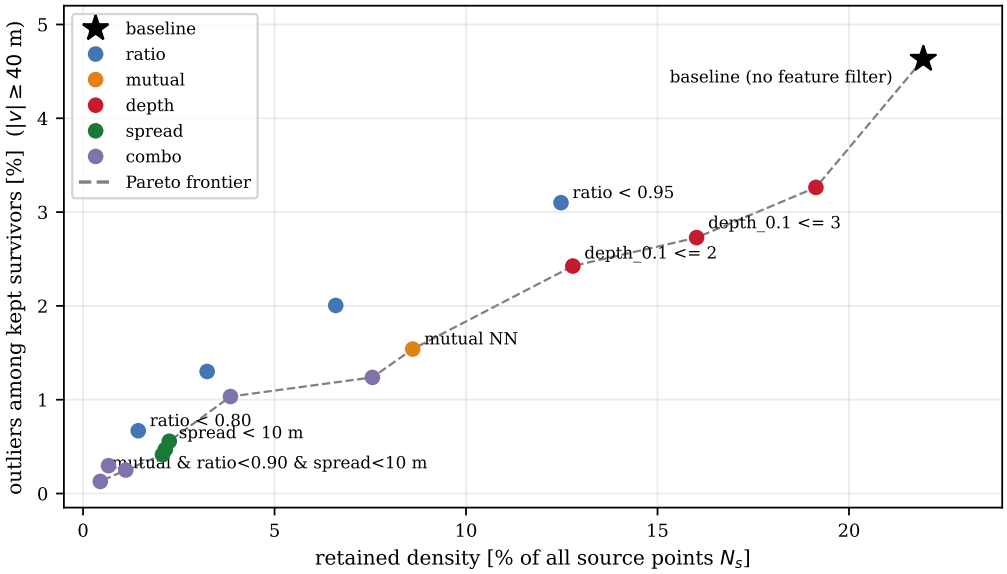


Figure 7: Density–purity operating points of all simulated rejection rules (composed with the supervoxel filter). The dashed line is the Pareto frontier. The spread family (green) occupies the low-density, very-low outlier corner; the depth family (red) traces the frontier at high density.

AUC is not filter quality. The spread metric ranks second-worst by AUC yet provides the cleanest practical filter. Threshold-free separability scores reward metrics that rank well *everywhere*; deployment only needs a metric that is trustworthy *somewhere* (here: the low-spread tail). Filter selection should therefore always include an operating-point analysis; in this study, an AUC-only protocol would have selected r_{10} and discarded the best filter.

7 Limitations

1. **Proxy labels.** All purity numbers rest on $\lvert v\rvert \geq \theta$; wrong matches with plausible magnitudes are uncounted, so purity values are lower bounds. Mitigated by bimodality and threshold invariance; removable only with reference data.

2. **Single tile, scene, and configuration.** All numbers describe one tile of the TCH sample clouds with GeDi-32 at 6 m radius. Generalisation to full-resolution clouds, other scenes, other descriptors (e.g. DIP-64), and other radii is untested.

3. **Censoring.** $\text{depth}_{0.2}$ is right-censored at $k = 20$.

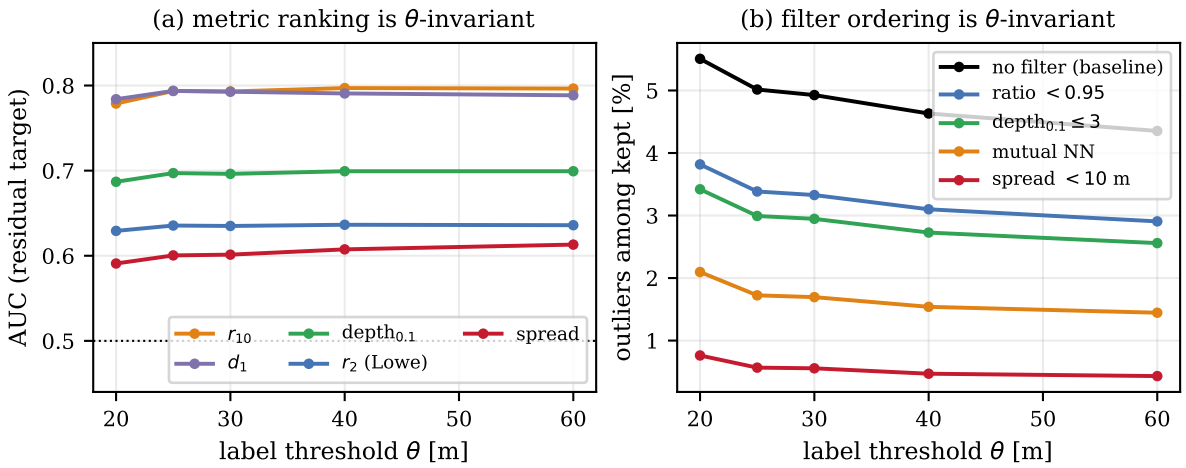


Figure 8: Sensitivity to the outlier-label threshold θ: (a) metric AUCs and (b) filter outlier rates vary smoothly and never change order for $\theta \in [20, 60]$ m.

4. **Unoptimised thresholds.** The filter grid is coarse; the true frontier may lie between grid points.

5. **Runtime cost unmeasured.** The spread filter needs a top-10 search and mutual NN a reverse search; the wall-clock cost of upgrading the correspondence stage from $k = 1$ has not been measured.

Missing for submission: Items to resolve before submission, in priority order: (1) replication on further tiles and the full-resolution epoch pair; (2) a reference dataset — identified stable (bedrock) areas or manually verified correspondences — to replace the proxy label for at least a subset of points; (3) bootstrap confidence intervals for Tables 2 and 3; (4) a k=100 re-run to de-censor $\text{depth}_{0.2}$; (5) runtime measurements for the $k \geq 10$ correspondence stage; (6) the DIP-64 descriptor as a comparison condition.

8 Conclusions and future work

On a rocky landslide with self-similar debris surfaces, the residual gross outliers of a deep-feature displacement pipeline are caused by deep, spatially global matching ambiguity — not by degenerate descriptors. The practical consequences are immediate: (i) a rejection filter belongs in the correspondence stage, exposed as an opt-in option; for a downstream user who densifies the field (e.g. RBF interpolation), the spread < 10 m rule offers an extremely clean skeleton (0.47% outliers), while $\text{depth}_{0.1} \leq 3$ preserves 73% of current density with 41% fewer outliers if the field is used as-is; (ii) evaluation protocols for such filters must report operating points, not only AUC. Future work: exploiting the spatial coherence of candidate sets at matching time, and training or fine-tuning descriptors for distinctiveness on self-similar natural terrain, which is where the information bottleneck demonstrably lies.

Missing for submission: Author contributions, acknowledgements, funding, and data/code availability statements.

References

[1] Z. Gojcic, C. Zhou, A. Wieser. F2S3: Robustified determination of 3D displacement vector fields using deep learning. *Journal of Applied Geodesy*, 14(2), 2020.

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[3] Y. A. Malkov, D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(4), 2018.

[4] D. G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2), 2004.

Missing for submission: Bibliography is limited to the four directly used sources; the related-work survey (Sec. 2) will add the remainder. Reference details (volume/issue/pages) should be verified — they were filled from memory, not from the publishers' records.